

▼ **Figure 56.6** **Habitat fragmentation in the foothills of Los Angeles.** Development in the valleys may confine the organisms that inhabit the narrow strips of hillside.



isolated patches of forest. Many other natural habitats have also been fragmented by human activities (**Figure 56.6**).

Habitat fragmentation often reduces the number of species in a region because the smaller populations in habitat fragments have a higher probability of extinction. Prairie covered about 800,000 hectares of southern Wisconsin when Europeans first arrived in North America but occupies only 800 hectares today; most of the original prairie in this area is now used to grow crops. Plant diversity surveys of 54 Wisconsin prairie remnants conducted in 1948–1954 and 1987–1988 showed that the remnants lost 8–60% of their plant species in the time between the two surveys.

Habitat loss is also a major threat to aquatic biodiversity. About 70% of coral reefs, among Earth’s most species-rich aquatic communities, have been damaged by human activities. At the current rate of destruction, 40–50% of the reefs, home to one-third of marine fish species, could disappear in the next 30 to 40 years. Freshwater habitats are also being lost, often as a result of the dams, reservoirs, channel modification, and flow regulation now affecting most of the world’s rivers. For example, the more than 30 dams and locks built along the Mobile River basin in the southeastern United States changed river depth and flow. While providing the benefits of hydroelectric power and increased ship traffic, these dams and locks also helped drive more than 40 species of mussels and snails to extinction.

Introduced Species

Introduced species—sometimes called invasive, non-native, or exotic species—are those that humans move intentionally or accidentally from the species’ native locations to new geographic regions. Human travel by ship and airplane has accelerated the transplant of species. Free from the predators, herbivores, pathogens, or competitors that limit their populations in their native habitats, such transplanted species may spread rapidly through a new region.

▼ **Figure 56.7** **Kudzu, an introduced species, thriving in the southeastern United States.**



Some introduced species disrupt their new community, often by preying on native organisms or outcompeting native organisms for resources. For example, the brown tree snake was accidentally introduced to the island of Guam from other parts of the South Pacific after World War II, as a “stowaway” in military cargo. Since then, 12 species of birds and 6 species of lizards that the snakes ate have become extinct on Guam. In 1988, the devastating zebra mussel, a suspension-feeding mollusc, was discovered in the Great Lakes of North America, most likely introduced in the ballast water of ships arriving from Europe. Zebra mussels form dense colonies and have disrupted freshwater ecosystems, threatening native aquatic species. They have also clogged water intake structures, causing billions of dollars in damage to domestic and industrial water supplies.

Humans have deliberately introduced many species with good intentions but disastrous effects. An Asian plant called kudzu, which the U.S. Department of Agriculture once introduced in the southern United States to help control erosion, has taken over large areas of the landscape there (**Figure 56.7**). The European starling was brought intentionally into New York’s Central Park in 1890 by a citizens’ group intent on introducing all the plants and animals mentioned in Shakespeare’s plays. It quickly spread across North America, where its population now exceeds 100 million, displacing many native songbirds.

Introduced species are a worldwide problem, contributing to approximately 40% of the extinctions recorded since 1750 and costing billions of dollars each year in damage and control efforts. There are more than 50,000 introduced species in the United States alone.

Overharvesting

The term *overharvesting* refers generally to the harvesting of wild organisms at rates exceeding the ability of their populations to rebound. Species with restricted habitats, such as those that inhabit small islands, are particularly vulnerable to overharvesting. One such species was the great auk, a large, flightless seabird found on islands in the North Atlantic. By